

Approach to Brief Resolved Unexplained Events (BRUE, formerly Apparent Life-Threatening Events)

Definition of brief resolved unexplained events (BRUE)

Brief resolved unexplained events (BRUEs) are defined as events occurring in infants < 1 year old that are characterized by the observer as:

1. **Brief:** < 1 minute but typically 20-30 sec
2. **Resolved:** patient returned to baseline state of health after the event
3. **Unexplained:** not explained by an identifiable medical condition
4. One of the following episodes:
 - a. Cyanosis or pallor
 - b. Absent, decreased or irregular breathing
 - c. Marked change in tone (hypertonia or hypotonia)
 - d. Altered level of consciousness

This should be in conjunction with a reassuring history, physical examination and vital signs at the time of clinical evaluation.

Approach to BRUE

A broad range of disorders can present as an apparent life-threatening event (ALTE)/BRUE (new terminology). A careful history and physical examination are essential to characterize the event, assess the risk of recurrence, and determine the presence of an underlying disorder. If the clinical history or physical examination reveals abnormalities, the patient may be at higher risk and further evaluation should focus on the specific areas of concern, such as possible non-accidental injury.

Important questions:

1. History of event
 - a. Who witnessed the event?
 - b. Where did the event occurred?
 - c. What was the child doing immediately before the event?
 - d. What was the state of the child during the event? (Colour change, tone, consciousness)
 - e. What was the state of the child after the event?
 - f. How long did the event last?
 - g. Is there any preceding illness/injury? Any drug history?
2. Past medical history
 - a. Birth history including gestational age
 - b. Symptoms of gastroesophageal reflux disease
 - c. Growth and developmental progress
 - d. Recent immunization
 - e. Family history of sudden unexplained death, sibling death/life-threatening event
3. Social history
 - a. Environmental history
 - b. Family set-up
 - c. Exposure to smoke, drugs

Causes of BRUEs

Cardiopulmonary

Channelopathies (e.g. Long QT syndrome, Brugada syndrome, catecholaminergic polymorphic ventricular tachycardia)
Ventricular pre-excitation (Wolff-Parkinson-White syndrome)
Cardiomyopathy (hypertrophic cardiomyopathy, dilated cardiomyopathy, myocarditis)

Child abuse

Abusive head injury
Shaken baby syndrome

Neurological

Epilepsy
CNS malformations

Infectious

Sepsis (bacteremia, urosepsis, meningitis)
Pertussis infection
RSV infection

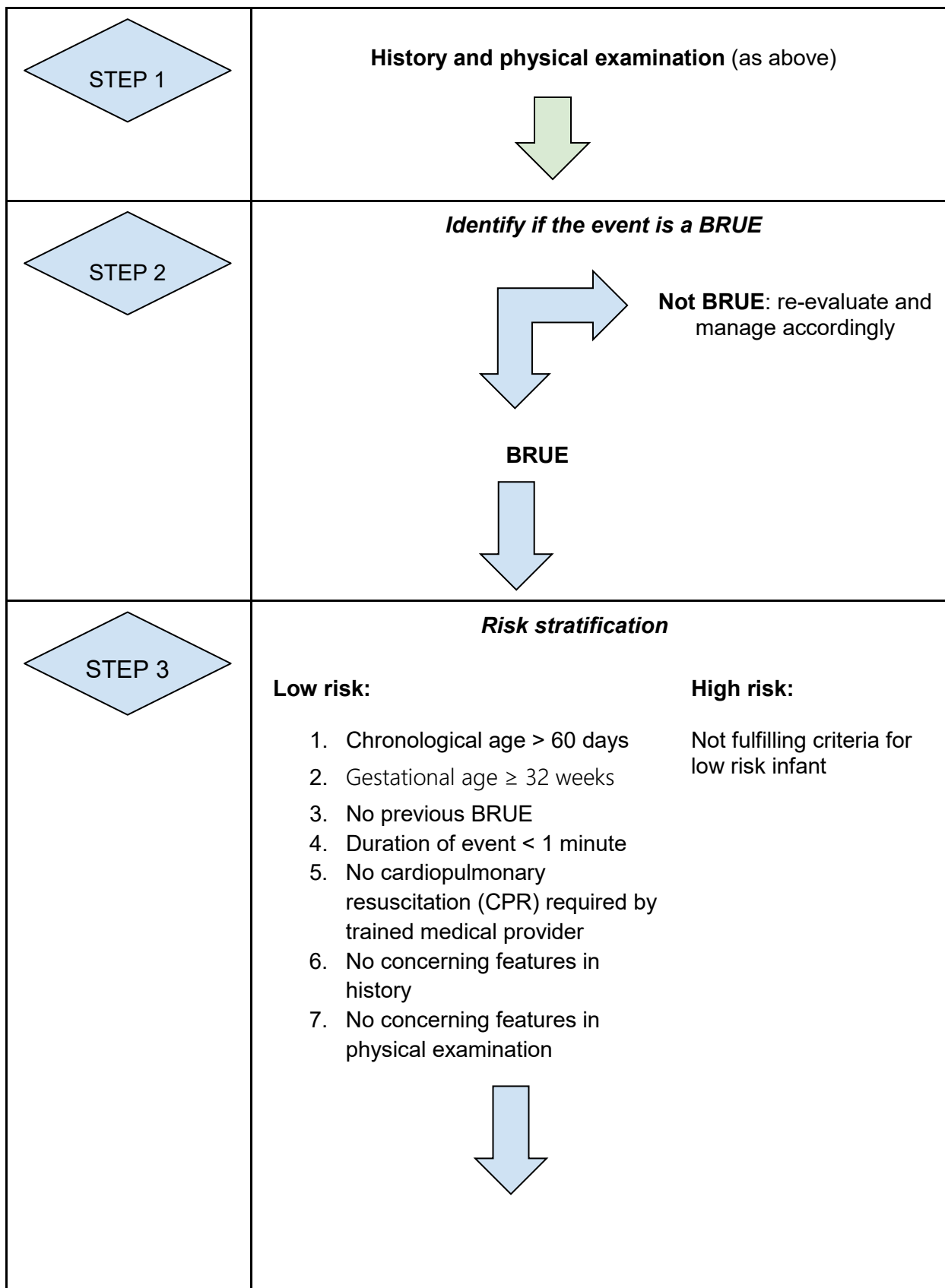
Aerodigestive

Gastroesophageal reflux
Aspiration

Metabolic

Electrolyte disturbances (e.g. hypocalcemia, hypoglycaemia, hyponatremia, acidosis)
Inborn errors of metabolism (e.g. urea cycle defects, organic acidemias, lactic acidemias, fatty acid oxidation defects such as medium chain acyl coenzyme A dehydrogenase deficiency)
Anemia

Management Algorithm



 STEP 4	Baseline investigations 1. 12-lead ECG 2. Continuous pulse oximetry for 1-2 hours 3. CONSIDER Nasopharyngeal swab for Bordetella pertussis PCR (turnaround time 1-3 working days, cost SGD \$115). Please discuss with the CE senior before proceeding. <i>Pertussis infection has been reported to cause apparent life threatening events in infants, as it causes gagging, gasping and colour change followed by respiratory pause. Such infants can be afebrile and may not develop cough or lower respiratory symptoms for several days afterwards.</i> <i>Decision to test a low-risk BRUE patient should consider potential exposures and vaccine history.</i> There is no need for (unless clinically indicated): 1. Full blood count, blood culture, CSF studies 2. Urinalysis (either bagged or catheterized sample) 3. Respiratory immunofluorescence 4. Chest X-ray 5. Blood gas analysis, serum lactate, bicarbonate, renal panel, glucose 6. Urine organic acids, plasma amino acids, plasma acylcarnitines 7. Neuroimaging (computed tomography, MRI or ultrasonography) 8. Electroencephalography (EEG) 9. Investigations for GER (upper GI series, pH probe study, barium meal) 10. Empirical acid suppression treatment for GER
 STEP 5	Disposition Patients who satisfy the above low-risk criteria and have normal ECG and continuous pulse oximetry may be discharged with early follow-up to General Paediatrics clinic in 2 to 3 days to trace the nasopharyngeal Bordetella pertussis PCR result. For the high-risk infant, proceed as per clinical suspicion

Reference

1. Tieder et al. Brief Resolved Unexplained Events (Formerly Apparent Life-Threatening Events) and Evaluation of Lower-Risk Infants. Pediatrics. 2016 May;137(5).